

Hoseong Jung

#Seoul National University, #Seoul, #Republic of Korea
ghtjdaleka@snu.ac.kr — +82 (010) 6616-5707 — LinkedIn — Webpage

RESEARCH INTERESTS

#Multi-Agent System, #Reinforcement Learning, #Learning-based Robot Control

EDUCATION

#Seoul National University

Ph.D. student in Interdisciplinary Program in Artificial Intelligence
B.S., Electrical and Computer Engineering

Sep. 2024 - Present
Mar. 2017 - Feb. 2021

RESEARCH EXPERIENCE

Seoul National University

Ph. D. Student

Seoul, Republic of Korea
Sep. 2024 - Present

- **Project: Intelligent Robot Navigation with Efficient LLM Interaction, Long-Term Consistency, and Location-Specific Expertise**
- **Project: Multi-UAV Air Combat Decision-Making and Maneuvering Algorithms for MUM-T**

Agency for Defense Development - AI & Autonomy Center

Fulltime Research Engineer

Daejeon, Republic of Korea
Jun. 2021 - May. 2024

- **Project: AI for Air Combat Engagement**
Developed simulated air combat environment for evaluating AI solutions
Researched reinforcement-learning-based policies for autonomous air combat
- **Project: Research on AI-based Context-Adaptive Swarming Unmanned Aircraft Operation**
Explored offline reinforcement learning approaches to create an autonomous air combat model

Research Officers for National Defense - Korean Navy

First Lieutenant

Jun. 2021 - May. 2024

Undergraduate Intern

LG Electronics

Jun. 2019 - Aug. 2019

AIoT Lab, Seoul National University

Oct. 2020 - Feb. 2021

PUBLICATIONS

#International

- **DynaFLIP: Rethinking Robotics Perception via Tri-Modal-Dynamics Guided Representation**
*Jusuk Lee, Seungjae Lee, Jonghun Shin, **Hoseong Jung**, Sungha Kim, Daesol Cho, H. Jin Kim, Jia-Bin Huang and Furong Huang*, in *Under Review*, 2026.
- **Cooperative Multi-Agent Reinforcement Learning via Epigraph-Form Guided Exploration**
Sungil Son, **Hoseong Jung***, Dahyun Oh*, H. Jin Kim*, in *Under Review*, 2026.
- **LAMP: Lane-Aligned Motion Primitives for Feasible Trajectory Prediction**
Sangjin Han, **Hoseong Jung***, Jeongtae Her*, Changhyun Choi, and H. Jin Kim*, in *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2026.
- **Temporal Action Representation Learning for Tactical Resource Control and Subsequent Maneuver Generation**
***Hoseong Jung**, Sungil Son, Daesol Cho, Jonghae Park, Changhyun Choi and H. Jin Kim*, in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. *Dahyun Oh, Yongjae Lim, **Hoseong Jung**, Jeongtae Her, Jusuk Lee, Changhyun Choi, H. Jin Kim and Jungwon Park*, in *International Journal of Control, Automation and Systems*, 2025.
- **Maneuver-Conditioned Decision Transformer for In-Flight Decision-Making**
***Hoseong Jung**, Yong-Duk Kim and Youngjung Kim*, in *IEEE Robotics and Automation Letters*, 2024.
- **Deep Reinforcement Learning-Based Air-to-Air Combat Maneuver Generation in a Realistic Environment**
*Jung Ho Bae, **H. Jung**, Seogbong Kim, Sungho Kim and Yong-Duk Kim*, in *IEEE Access*, 2023.

#Domestic

- **AI-based Air-to-air Close Combat Model**
*Y. -D. Kim, S. Kim, J. H. Bae, **H. Jung**, S. Kim*, in *Avionics Systems Symposium Korea*, 2022. **Best Paper Award**

SKILLS

- **Programming:** # Python; C++;
- **Software & Package:** # Tensorflow; Pytorch; Ray;

PATENTS

- **Operating Method for Electronic Apparatus for Configuring Artificial Intelligence Model and Electronic Apparatus Supporting Thereof**
Hoseong Jung, Youngjung Kim, Yong-Duk Kim, registration number: 10-2694807, *KR Patent*
- **Learning Method for Computer Generated Force Capable of Adjusting Engagement Level and Electronic Apparatus Therefor**
Hoseong Jung, Yong-Duk Kim, Sungho Kim, Minseok Kim, Sangmin Lee, Yongwon Lee, registration number: 10-2620631, *KR Patent*
- **Learning Method for Aircraft Control and Electronic Apparatus Thereof**
Jung Ho Bae, Hoseong Jung, Sungho Kim, Yong-Duk Kim, registration number: 10-2617822, *KR Patent*
- **Learning Method for Controlling Unmanned Aerial Vehicle and Electronic Device for Performing the Same**
Jung Ho Bae, Seogbong Kim, Sungho Kim, Yong-Duk Kim, Insu Hwang, Hoseong Jung, registration number: 10-2559608, *KR Patent*